

The empirical recurrence for OEIS A250679: recovering a conjecture that is not written down

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Abstract

OEIS A250679 records “Empirical recurrence of order 58”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 96$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 58, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 58 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A250679 is “Number of $(3+1)X(n+1)$ 0..3 arrays with nondecreasing $x(i,j)+x(i,j-1)$ in the i direction and nondecreasing absolute value of $x(i,j)-x(i-1,j)$ in the j direction.”. It has offset 1 and begins

2512, 34630, 388916, 4008211, 38224684, 345255872, 2996016017, 25227877364, ...

The entry states:

Empirical recurrence of order 58 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:40:50, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 96$ of the 256, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 96$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 192$ exact terms of the model returns order 58 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{58} c_i a(n-i),$$

c_1	25	c_{16}	−79900602078	c_{31}	126117217289612	c_{46}	168659147495936
c_2	−155	c_{17}	−585723316094	c_{32}	−2178930488445715	c_{47}	−68223135978496
c_3	−1034	c_{18}	794015476714	c_{33}	118990560849959	c_{48}	−48204115765248
c_4	14855	c_{19}	2978171829092	c_{34}	2820058999043339	c_{49}	18298802671616
c_5	−9457	c_{20}	−5245505286874	c_{35}	−472376081290566	c_{50}	10617872691200
c_6	−514579	c_{21}	−11797788308690	c_{36}	−2978525389769971	c_{51}	−3550381424640
c_7	1529552	c_{22}	25672883460122	c_{37}	757851527306085	c_{52}	−1744595615744
c_8	9588501	c_{23}	36497534854904	c_{38}	2564708411632015	c_{53}	469897216000
c_9	−47544477	c_{24}	−97280739093819	c_{39}	−821077634982592	c_{54}	201993682944
c_{10}	−102183733	c_{25}	−87471526031341	c_{40}	−1795684485380920	c_{55}	−37890293760
c_{11}	843967558	c_{26}	292014501754843	c_{41}	659660234594920	c_{56}	−14736162816
c_{12}	473802999	c_{27}	158166221362614	c_{42}	1017851852570928	c_{57}	1401421824
c_{13}	−10151686609	c_{28}	−703772098052833	c_{43}	−404388615506560	c_{58}	509607936
c_{14}	3184608777	c_{29}	−199868201599729	c_{44}	−464097882030336		
c_{15}	88772644316	c_{30}	1372996183473713	c_{45}	190265990435968		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 58, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $58 + 4$ terms removed returns order 58 as well, so $\deg q_{\min} = 58 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $58 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 58 that this sequence satisfies — and that family turns out to have exactly one member.

References

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